Keeping GEMM Results Block-Scaled across the Tensor–Vector Hand-off: When Does It Pay Off?

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***Abstract*—**[abs-p1] **A GEMM result can be consumed where it is produced, in a fused epilogue, or take a scalar or block-scaled form and pass to a separate vector unit. The choice changes where rounding happens, how long execution resources stay occupied, and how much intermediate state is stored and in what form. We study when keeping the result block-scaled across this matrix-to-vector hand-off pays off, in a processing element (PE) pairing a systolic Tensor unit with two Vector cores [TBD-D1: form, width and division of labor of the two Vector cores]. We compare five placements of the hand-off: up to four as compile-time configurations of one RTL, the fused one with a synthesis slice and a calibrated cycle-level model; numerical error is evaluated on synthetic inputs. We characterize where this pays off by fusion feasibility, result lifetime and metadata transaction granularity [TBD-E4: where the block form pays off and where it does not], and report the cost of a fail-visible hand-off [TBD-E8: implementation cost] and a system-level projection [TBD-P: system-level result; source of performance data]. Decoding with three public models from two generations of one family (DeepSeek-V4-Pro and V4-Flash, then V4.1-Flash) motivates the design and is the evaluation workload. Today's RTL prototype has one Vector core and implements one placement; physical results are post-route EDA estimates on the ASAP7 predictive PDK, not silicon measurements.**

I. Introduction

[sec1-p1] The result of a general matrix multiplication (GEMM) can be consumed where it is produced, in an epilogue fused into the matrix engine, or it can take a scalar or a block-scaled form and be handed to a separate unit. The choice changes where values are rounded, how long execution resources stay occupied, and what it costs to store intermediate results. This paper studies how these effects together decide when the block form is worth keeping.

[sec1-p2] Consider a GEMM followed by a residual add (Fig. 1). In B-D (block, direct), the matrix engine's outlet forms a block-scaled result and a vector unit adds the residual in the block domain. In S-D (scalar, direct), the outlet converts each value to scalar floating point. In S-E (scalar, fused epilogue), the producer adds the residual in its epilogue before forming the output representation. As quantization does not commute with the add ($Q(A) + R \neq Q(A + R)$ in general), equal mantissa widths do not give equal numerical quality; each placement's computation and rounding order must be frozen before costs are compared. Removing a scalar conversion does not guarantee lower cost, and fusion, which saves a transfer, can hold accumulators longer or load outlet ports.

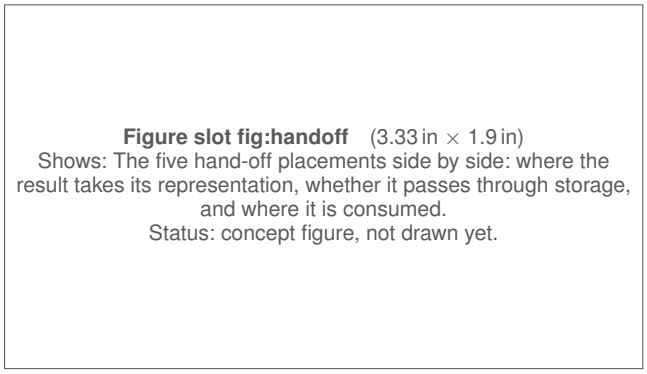


Fig. 1. Five hand-off placements. A block-scaled (B) or scalar (S) result crosses directly (D) or through storage (M); S-E instead fuses consumption into the producer's epilogue.

[sec1-p3] In the production designs we examined whose public material suffices to determine the hand-off format, the matrix engine passes scalar values to vector execution, block formats are rebuilt by dedicated quantizers, and vector instruction sets accept a block scale only in conversion instructions (Section II-B). Academic designs carry block formats across the matrix/non-matrix operator boundary [17] [ref: MXInt ViT] [ref: MASE], so the crossing itself is not new (Section VII). What remains open is where a processing element (PE) whose Tensor unit and Vector cores are separate, reused resources, with limited storage and ports, should form the block representation and how long to keep it.

[sec1-p4] The motivating workload is decoding with three public models from two generations of one family: DeepSeek-V4-Pro and V4-Flash, then V4.1-Flash [7]–[9] [ref: DeepSeek-V4 paper] [ref: DeepSeek-V4-Flash model repository]. From V4 to V4.1, each model component's share of a decode step, the storage formats and the scale-block sizes changed, while a set of structural quantities held (Section II-D). These changes motivate the target design; the models' analytically derived operation counts form the evaluation workload (Section V-D).

[sec1-p5] A block's payload means nothing without its scale; a hand-off is fail-visible if no payload can be consumed under

Working draft (first full text, 2026-10-06) under a working title; not peer reviewed, not submitted to any venue. Colored brackets mark open items; gray tags name paragraphs; empty frames stand for figures to come. Sources: https://github.com/fifakaka35357/ScaleNative

stale or mismatched metadata without a reported error. We ask one question in two parts. In a PE that combines a matrix unit with vector units, when does keeping GEMM results block-scaled across the matrix-to-vector hand-off—forwarded directly, or committed with their scales and re-consumed—pay off against a scalar floating-point hand-off and a fused epilogue at equal numerical quality? And what does it cost to make that hand-off fail-visible and to check scale metadata at dispatch time versus in the datapath? [TBD-N: final wording; whether to name transaction granularity]

[sec1-p6] We keep the forms of evidence apart. Up to four hand-off placements are compile-time configurations of one RTL, compared in RTL simulation. Producer-side fusion is not part of the PE RTL; we evaluate it with a synthesis slice and a cycle-level model, and report an optimistic bound and a calibrated estimate. Numerical error is evaluated on synthetic inputs by a model that follows the hardware's accumulation and rounding order. System-level impact is a model projection [TBD-P: source of cycle-level and system-level performance data]. The text describes the target design; each subsection of Section IV ends with what today's RTL prototype implements. Today that is one placement, B-D; the workload characterization and a post-route layout exist, and no placement has yet been compared.

[sec1-p7] We make two contributions. (1) A finding: when keeping the block representation across the matrix-to-vector hand-off pays off, characterized by whether fusion is feasible, how long a result must live, and the granularity of metadata transactions [TBD-E4: boundary of the pay-off region; no experimental result yet]. (2) An engineering mechanism: a fail-visible hand-off and commit protocol for block results, which reports mismatches and errors to the PE's command processor and publishes a result only when its payload and scale metadata are complete; it includes the fast lane, the direct link from the Tensor unit to the Vector cores. The RTL, its bit-accurate reference model and the evaluation models are the evidence platform, not separate contributions.

II. Background and Motivation

A. Block-Scaled Formats and Three Block Axes

[sec2.1-p1] A block-scaled format lets k elements P_i share one scale X; element i represents $X \cdot P_i$ [4]. We use the term for three families: block floating point (BFP), where integer mantissas share an exponent and the block size varies by design [5], [10], [11]; OCP MX, with $k = 32$ and an E8M0 scale [4], [6]; and NVFP4, with E2M1 elements, an E4M3 scale per 16 elements and an FP32 scale per tensor [1].

[sec2.1-p2] We keep three block axes apart; values are the prototype's. The input scale block runs along the inner-product dimension K: in the MAC cells' MX input mode, 32 elements share an E8M0 scale. The output block runs along the output dimension N: a row's 16 elements share one power-of-two exponent. The storage group is a row of the scale sidecar, which stores block exponents apart from the payload, 16 per row. The GEMM outlet thus forms a new block representation and passes no input scale through [TBD-D7: how the three axes read in the target design].

B. Where the Hand-off Happens Today

[sec2.2-p1] We survey the designs we examined whose public material suffices to determine the hand-off format. On NVIDIA Blackwell, block-scaled products accumulate in FP32 and cuBLAS quantizes at the end of the GEMM [13], [14]. MTIA 2i forwards FP32 results directly to a floating-point SIMD engine [3]. Tenstorrent stores BFP; its default element-wise path unpacks to FP32 or BF16 before compute and packs on write-back [15], [16]. Trainium3 and Maia 200 output scalars; block formats are rebuilt downstream [ref: Trainium3; Maia 200]. Outside matrix multiplication, Intel AMX, PTX and CDNA 4 accept a block scale only in format-conversion instructions [2], [14] [ref: Intel AMX].

[sec2.2-p2] We do not claim that these designs could not keep a block representation, and we do not dismiss prior work as special-purpose; we ask under which conditions the placement of the hand-off changes cost and numerics. Two surveyed practices are strong baselines: the fused end-of-GEMM quantization is the production counterpart of S-E, and direct matrix-to-vector forwarding is established practice, so B-D differs only in the forwarded format. What public material does not disclose is not assumed absent.

C. Two Hypotheses to Test

[sec2.3-p1] We pose two hypotheses, not conclusions. Hypothesis A: fusion placement, result lifetime and the point of numeric narrowing together decide whether a block-form hand-off pays off. For a local, single-consumer operation, producer-side fusion wins under some resource conditions, whereas a result that is retained, consumed repeatedly, or held for cross-block information may favor the block form in a predictable region. Hypothesis B: metadata transaction granularity can decide whether keeping a block form resident helps or hurts, with the numerical representation unchanged. We test them separately and combine them in Section VI-C, with narrowing points fixed per placement (Section III-A).

D. A Workload in Motion and the Design Questions It Raises

[sec2.4-p1] The three models share a context limit of one million tokens. We derive the multiply–accumulate operations (MACs) per decoded token analytically from their public configurations and reference implementations (Section V-D), and group them by model component: index scoring (an indexer's 128-dimensional inner products with cached keys, which select the top-k entries), the attention core (scores and weighted sum of the attended entries under an online softmax) and the feed-forward experts. The mix shifts with context (Fig. 2, Table I): at 8K tokens the feed-forward experts hold 51.3–55.2% of the MACs; at 1M index scoring is the largest component, at 38.3–64.3% (V4.1-Flash by its technical report's count, Section V-D).

[sec2.4-p2] Structural quantities held across the three models: head dimension 512, index dimension 128 and an online

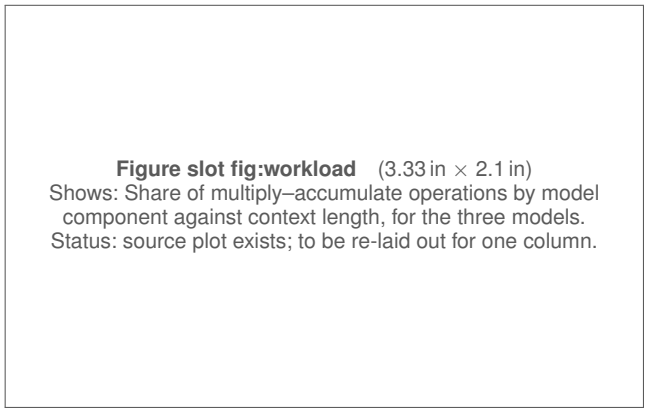


Fig. 2. Share of multiply–accumulate operations per decoded token, by model component, versus context length (analytical derivation). V4.1-Flash at 64K and above follows its report's count.

TABLE I
DEMANDS OF ONE DECODE STEP FOR THREE MODELS FROM TWO GENERATIONS; ANALYTICAL COUNTS, NOT MEASUREMENTS. V4.1-FLASH AT 1M FOLLOWS ITS REPORT'S COUNT; CACHE BYTES INCLUDE BLOCK SCALES.

Per decoded token	V4-Pro	V4-Flash	V4.1-Flash
Layers / hidden width	61 / 7,168	43 / 4,096	40 / 5,120
MACs at 8K context (10^{10})	5.466	1.477	1.796
MACs at 1M context (10^{10})	15.16	7.017	2.875
Largest at 8K: feed-forward (%)	51.6	51.3	55.2
Largest at 1M: index scoring (%)	42.5	64.3	38.3
Cache bytes read at 1M (MB)	705.3	479.3	183.3
Main key–value entry, 512 dims	448 8-bit + 64 BF16		512 4-bit
8-bit blocks: weights / activations	128×128 / 128		32×32 / 32

All three: one ReLU and weighted add per 128 index-scoring MACs; one softmax-accumulator rescale per 128 attention-core MACs.

softmax in groups of 64 entries. Operation shares, storage formats and scale-block sizes changed: V4.1-Flash stores its main key–value entries in 4-bit form and shrinks 8-bit blocks from 128 to 32. Its authors state that this format saves storage rather than speeding up matrix multiplication [9]; the V4 authors propose replacing SwiGLU with an element-wise activation free of exponentials and divisions, so that post-GEMM processing does not stall the GEMM pipeline [ref: DeepSeek-V4 paper].

[sec2.4-p3] Six design questions follow; the target design's answers are planning points, and Section IV states what the prototype has. (i) Which unit executes non-GEMM operations: two Vector cores per PE, their form open (Section IV-D). (ii) One PE type or two: one, with capability bits for a second. (iii) Compute per bit-width tier: 2048, 1024 and 256 MACs per PE per cycle at 4-bit, 8-bit and BF16. (iv) What a block-scaled hand-off saves: this paper's question. (v) Shared storage and PE port: 128–256 MiB per 8×8 mesh and a local port of 64 payload bytes per cycle into each PE (Section IV-A). (vi) Fixed cost per tile, one pass of the array: 5 cycles.

III. KEEPING RESULTS BLOCK-SCALED: DESIGN SPACE, REPRESENTATION AND PROTOCOL

A. Five Hand-off Placements

[sec3.1-p1] Table II defines five placements of the hand-off. B-D and S-D forward each output row directly to the Vector side, as one block or as scalars; S-E consumes the result in an epilogue at the producer; B-M and S-M first write it to L2, the PE's local storage, and read it back before consumption [TBD-D1: whether the placements are compared on one Vector core or on both].

[sec3.1-p2] Four constraints apply to all placements; one placement pays off over another only under them and at a common error budget [TBD-N: deciding metric and error budget, frozen before the sweep]. The primary comparison fixes the physical bit width and the total byte budget, with L2 at its target 128 KiB (32 KiB in the prototype); a secondary comparison fixes the number of blocks held. Schedules are matched, so no scheduling gain is credited to the representation. A conversion before the next GEMM is charged to every path needing it. Accumulation precision, residual injections, narrowing points, reduction order and rounding are frozen before any performance comparison; the target design's outlet defaults to round-to-nearest-even [TBD-D7: rounding at each narrowing point, to be re-read from the RTL once merged].

B. What Each Comparison Can Attribute

[sec3.2-p1] A pairwise comparison isolates the representation, materialization in storage or the fusion placement, each under a condition that it must establish; B-D versus S-E compares two complete designs (Table III). When B-D, S-D, B-M and S-M all have the same form of evidence, the interaction for a cost C is $I = (C_{B,M} - C_{S,M}) - (C_{B,D} - C_{S,D})$; costs backed by different evidence are never subtracted.

C. Consumer Tiers

[sec3.3-p1] We define three consumer tiers. The first is a residual add. The second is a per-block sum of squares. The third computes a statistic and then consumes the original result again, as a normalization does: with X the GEMM output accumulated over the full K, $s_r = \sum_{j \in \mathcal{J}_r} X_{rj}^2$ and $Y_{rj} = X_{rj}\, h(s_r)$, where h is a scalar function and $\mathcal{J}_r$ is one 16-element block or a row spanning at least two output blocks [TBD-E4: whether the third tier is evaluated, and at which sizes]. The representation of X and the cost of h are fixed in advance; if a host or a model computes h, every placement pays the same cost. The fused baseline may buffer the 16 values or replay them; no placement retains a result for free.

D. Retaining and Publishing a Result

[sec3.4-p1] An output transaction (a producer's output rows with their metadata) is published, that is, made visible to consumers outside it, only after all its rows and metadata are complete and no sticky error is set. A producer–consumer chain bound to the transaction may consume rows as they arrive, but the object it writes stays unpublished; a late error

TABLE II
THE FIVE HAND-OFF PLACEMENTS. B/S: BLOCK OR SCALAR REPRESENTATION; D/M/E: DIRECT, THROUGH STORAGE, OR FUSED EPILOGUE.

	Representation	Through storage	Numeric order; N1, N2: narrowing points	Consumer and resources	Evidence
B-D	one block per row: 16 mantissas of 16 bits, shared power-of-two exponent	no: FIFO, then ingress check	acc. → N1: block formed, 48 to 16 bits, row's largest exponent → block-domain residual add → N2 → write	16 lanes of 16-bit mantissas, one exponent per block [TBD-D1: one core or both]	RTL simulation
S-D	scalar BF16 or FP32, as separate groups	no	acc. → N1: outlet converts to BF16 or FP32 → scalar residual add → write	scalar lanes; converters replace block formation	RTL simulation
S-E	FP32 in the producer; output formed after the epilogue	no; residual read from L2	acc. → FP32 residual add in the epilogue → N1: output formed	FP32 epilogue; may buffer 16 values or replay	synthesis slice, cycle-level model
B-M	block: payload plus scale sidecar	yes: write, read back, consume	as B-D up to the write; stored without narrowing; block-domain consumption	as B-D; scale sidecar update is a read-modify-write	RTL simulation
S-M	true BF16 or FP32 encoding	yes	acc. → N1: to BF16 or FP32 → write → read back → scalar consumption	as S-D; encoder at write, decoder at read	open [TBD-E4: RTL or model]

acc.: accumulation over the full K. Target design with the prototype's block size and bit widths (Section II-A); only B-D is in the prototype (Section IV-F).

TABLE III
ATTRIBUTION MATRIX: WHAT EACH PAIRWISE COMPARISON ISOLATES, AND THE CONDITION IT MUST ESTABLISH, NOT ASSUME.

Pair	Isolates	Condition to establish
B-D/S-D	representation	equal resources, numeric contract, capacity
B-M/S-M	representation	round trip, same protection, true encoding
B-D/B-M	storage	both complete the same workload
S-D/S-M	storage	BF16 and FP32 as separate groups
S-D/S-E	fusion placement	equal accumulation precision, rounding points, consumer precision, final quantization
B-D/S-E	whole designs	isolates neither conversion nor fusion

Figure slot fig:retain (3.33 in × 2.6 in)
Shows: Result retention and publish path, with the protocol as one panel.
Status: only a sketch exists.

Fig. 3. Retention and publish path of one result (target design), with its three separate events. The prototype implements the path up to payload write-back.

is attributed to the original transaction and blocks publication. Snapshot reuse, credit return and transaction retirement are three separate events (Fig. 3) [TBD-D7: layout of the written object, pending the write-back design].

[sec3.4-p2] One constraint shapes this path: the PE's command processor releases a context, one of its command queues, when the GEMM is done, before its output has drained. By our reading of the RTL (not simulated), deferring the release until the output retires would deadlock a chain that alternates producer and consumer commands in one context: the consumer could not be dispatched, the hand-off FIFO would fill, the outlet would stall and the GEMM would never complete. Resources are released early, and a completion record must outlive them to settle rows in flight.

E. Objects, Lifetime and In-flight Identity

[sec3.5-p1] The target protocol tracks three kinds of information. The numeric encoding of an object (its scales, codec, grouping and valid range) is stored with the object rather than rebuilt from what the consumer expects. Its lifetime (which version is visible, who holds read leases, who may write) lives in a bounded object table in the command processor. The identity of a result in flight travels with every capsule (one output row with its metadata): context, tag, row number and a last-row mark [TBD-D7: where the encoding is stored, when a lease ends].

F. Metadata Transaction Model

[sec3.6-p1] Hypothesis B (Section II-C) is tested with a transaction account. With w bits per physical scale sidecar access, k elements per block and g blocks covered by one merged update, a read-modify-write costs $C_{\mathrm{RMW}} = 2w/(kg)$ bits per element; g is bounded by buffer capacity, arrival order and commit time. The SRAM macros have no byte mask, so every partial update is a read-modify-write that can collide with read requests. The account has five more terms (payload accesses, tail blocks and masks, object-table accesses, merge waiting, conflict stalls) and uses one production-to-consumption boundary for B-M and S-M. Ordinary write combining is the strong baseline [TBD-D7: layout of the scale metadata, pending the write-back design].

IV. MICROARCHITECTURE

A. PE Organization and Evaluation Boundary

[sec4.1-p1] PEs of one type form a mesh whose links are the on-chip network (NoC), over which a mesh-level Control Core dispatches work (Fig. 4). A PE holds one Tensor unit, an output-stationary systolic array of 16×16 cells, and two Vector cores (Fig. 5). The Tensor and Vector sides share L2 (128 KiB)

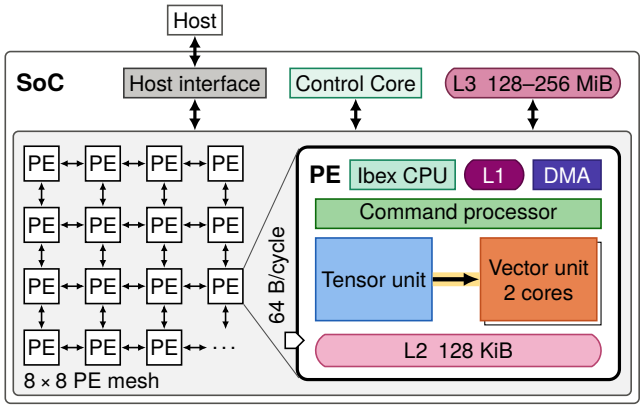


Fig. 4. System organization of the target design: a mesh of identical PEs, whose links are the on-chip network, with a host interface, a Control Core and a shared L3. One PE is enlarged; the thick arrow is the Tensor-to-Vector hand-off. The prototype RTL is one PE with a single Vector core and a 32 KiB L2; nothing above the PE is implemented.

and a command processor but no arithmetic; an Ibex RISC-V CPU [ref: Ibex], which has its own L1 memories, feeds the command processor with descriptors. The local port into each PE (the pentagon in Fig. 4, which draws the inbound direction only) is planned at 64 bytes of payload per cycle [TBD-D4: channels and handshake of the PE port, pending the network design]; shared storage at the mesh edge (L3) is planned at 128–256 MiB per 8×8 mesh [TBD-D3: bandwidth from shared storage into the mesh]. The mesh size is a planning point, and where the host interface, the Control Core and L3 attach to the mesh is not fixed. Where the design is open, Fig. 5 is generic: its two Vector cores are drawn equal, the outgoing port has no width, and the link from Vector core 1 to L2 is dotted.

[sec4.1-p2] The evaluation boundary is the PE interface. Inside it, the PE is evaluated in RTL and the fused epilogue as a synthesis slice with a cycle-level model. Everything outside is modeled: on-chip network, shared storage and off-chip memory [TBD-D3: whether network endpoints and shared storage enter RTL]; Fig. 4 draws neither this boundary nor off-chip memory. *Prototype status:* one Vector core, 32 KiB of L2, a 32-bit stream port, no RTL outside the PE; run-time errors never reach the command processor.

B. Tensor Producer

[sec4.2-p1] Each cell of the array (Fig. 6) is a fused multi-precision MAC built from 4×4-bit multiplier primitives, with the prototype's input modes (one BF16, one MXINT8, two FP8 or four FP4 operand pairs per cycle), a wide floating-point accumulator and a 61-bit snapshot. The target design performs 2048, 1024 and 256 multiply–accumulates per PE per cycle in its 4-bit, 8-bit and BF16 tiers [TBD-D5: which input modes map to each tier]. The target rates, printed in Fig. 6, are nominal, and the BF16 one is provisional. The figure is generic where the design is not fixed: the cell interior, the number of output banks, the tile-controller connection, the preload source and escape handling.

[sec4.2-p2] A row scan forms the output block at the outlet: it takes the row's largest exponent and narrows 48-bit mantissas to 16 bits, the first narrowing point; a tile yields 16 capsules. A hard escape (NaN or infinity) ends the transaction in error; a soft escape (underflow within a block) is consumed and counted by default. The fixed cost per tile is planned at 5 cycles [TBD-D6: timing of this overhead, not yet evaluated]. *Prototype status:* 56 fixed cycles per tile by the tile controller's state sequence (not simulated), after operands are moved into the Tensor unit's operand buffer; escape rows are dropped.

C. Tensor-to-Vector Hand-off and Fail-Visibility

[sec4.3-p1] The fast lane carries each output row from the array outlet to a Vector core as one capsule: a 256-bit payload plus metadata (block size, scale format, scale epoch, context, tag, row number and a last-row mark). It is a one-way valid/ready FIFO whose ready signal gates the row scan, so a slow consumer back-pressures the producer. Each of the two outlet ports has its own row scan, block formation and fast lane [TBD-D1: buffering, credit return and check position of the two fast lanes].

[sec4.3-p2] Each capsule is checked at ingress, ahead of the consumer's operand queue: block size, scale format, scale epoch and identity must match what the consuming command expects. Any mismatch is fail-visible: it reaches the command processor with its own error type, is attributed to the original transaction and blocks publication. Checking scales at dispatch instead, against the object table of Section IV-E, is a candidate alternative. *Prototype status:* one fast lane, with three format checks only; a mismatch stalls the consumer until a timeout near 1023 cycles.

D. Vector Consumer and the Second Hop

[sec4.4-p1] The prototype's Vector core computes in block floating point: 16 lanes of 16-bit mantissas under one exponent per block. It provides residual add, rescale, in-block maximum and sum, a sum of squares in a 40-bit window, and packing; the target design (Fig. 7) adds index-score post-processing and top-k selection. The figure shows core 0 at block level and core 1 folded; its box sizes do not indicate compute, and operands read back from L2 pass the same ingress check. The PE must stay fully functional with one Vector core masked; the candidates are a pipelined plus a general-purpose core, or two identical cores [TBD-D1: form, width and division of labor of the two Vector cores].

[sec4.4-p2] The second hop writes the result to L2 or feeds the next GEMM. A write renormalizes it, the second narrowing point, and commits payload plus block exponents in the scale sidecar for later re-consumption. A feed first converts it to that GEMM's input format. *Prototype status:* write-back stores only the payload, so neither the round trip nor the path into a next GEMM closes; no command can yet invoke the sum of squares.

E. L2, Commit Protocol and Control

[sec4.5-p1] L2 obeys one contract for its behavioral model and single-port macros: a bank serves one access at a time and

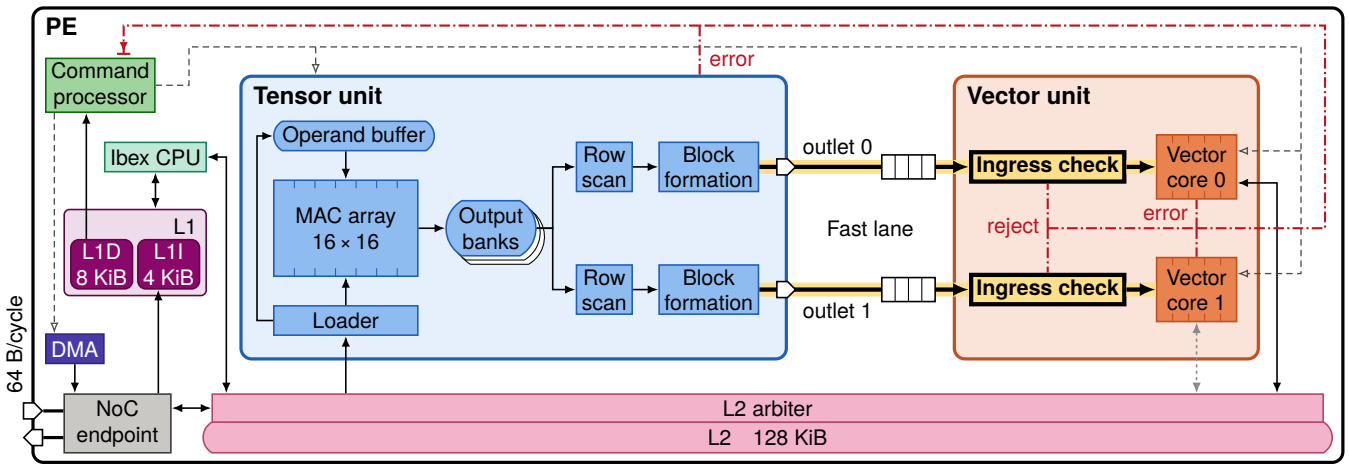


Fig. 5. PE microarchitecture (target design). Thick lines: the hand-off path; dash-dotted red: errors and rejects; dotted: undecided; stack and queue symbols imply no counts or depths. Among the differences of today's prototype RTL: one Vector core and one fast lane, a 32 KiB L2 that the DMA engine accesses directly, and execution-time errors that do not yet reach the command processor.

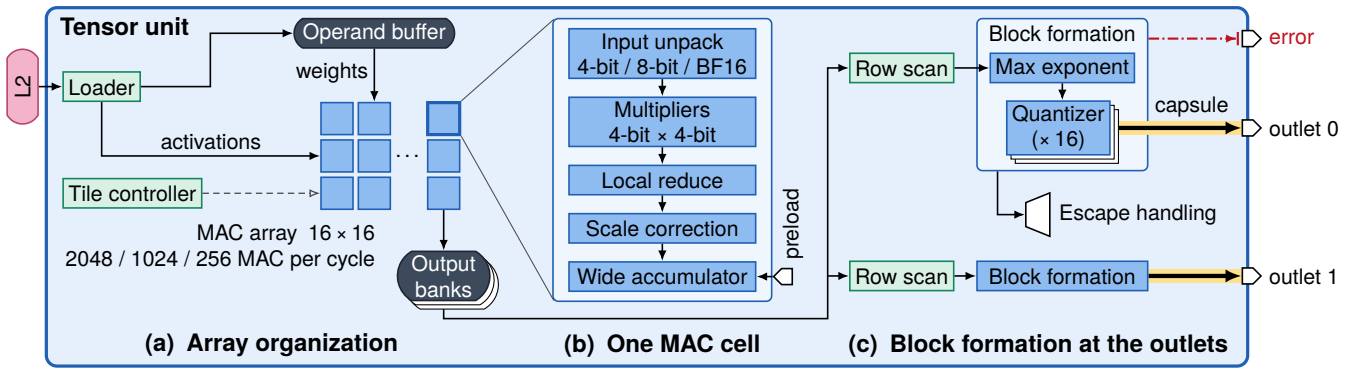


Fig. 6. Tensor unit of the target design; thick lines mark the capsule hand-off and the dash-dotted line the error path to the command processor. The prototype has one outlet and one output bank, loads both operands before each tile, performs 1024 / 512 / 256 multiply-accumulates per cycle, and lacks preload, escape selection and the error path.

acknowledges only requests it served, and sub-word updates (a scale sidecar entry) execute atomically. A commit writes payload and scales; the command processor records the object in a bounded table, grants read leases and publishes it once the transaction retires error-free. The command processor rejects illegal descriptors at fetch, dispatches by per-unit credits and settles each command's state and credit exactly once [TBD-D7: L2 contract and write-back design open]. *Prototype status:* with macros a conflicting read can return stale data marked valid; no object table, leases or publication.

F. One RTL, Several Hand-offs

[sec4.6-p1] Up to four placements are compile-time configurations of one RTL, differing only at the hand-off. S-D swaps block formation for outlet converters and scalar lanes; B-M adds scale write-back and read-back; S-M adds scalar storage [TBD-E4: whether the scalar-through-storage placement is built in RTL or modeled]. *Prototype status:* only B-D exists.

TABLE IV
CONFIGURATIONS AND FORMS OF EVIDENCE.

Configuration	Definition
PROTOTYPE	RTL as it stands; behavioral dual-port L2 in simulation
CORRECTED	plus a versioned list of fixes; main experiments
PROJECTED	plus more PEs, network and shared storage; projections
(listed apart)	target-design points not yet in RTL

Forms of evidence: RTL simulation, cycle-level model, analytical derivation, synthesis slice, post-route EDA estimate.

V. EVALUATION METHODOLOGY

A. Configurations and Evidence Types

[sec5.1-p1] Every number carries two labels: one of three configurations [TBD-N: neutral names for the three configurations] and a form of evidence (Table IV, Fig. 8). The fixes of CORRECTED include error reporting to the command processor, the contract of the single-port L2 macros and scale write-back [TBD-D7: final list of fixes, pending the merge plan]. A physical result is never called measured.

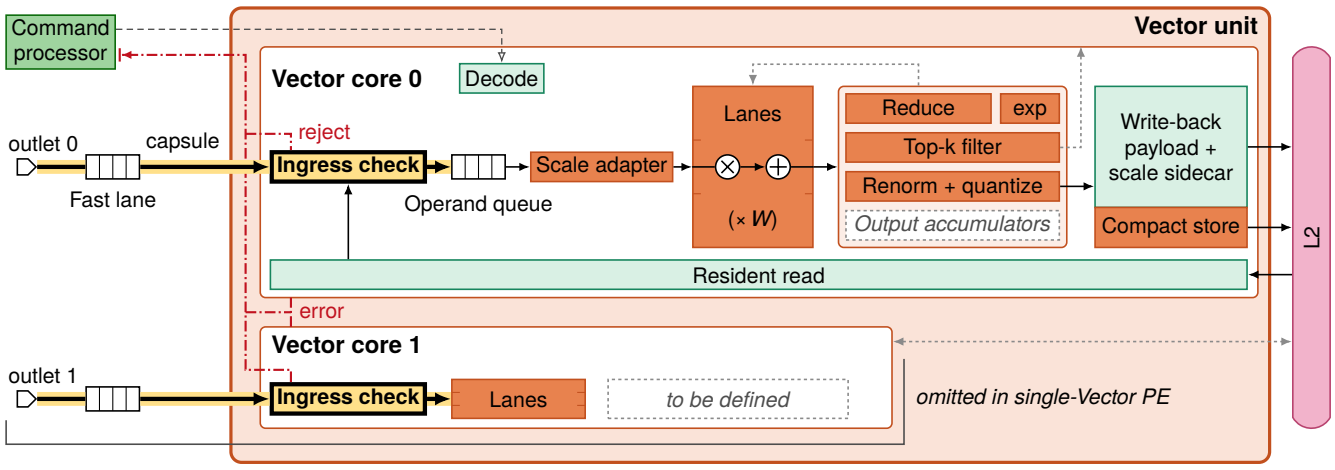


Fig. 7. Vector unit of the target design: each Tensor outlet feeds one Vector core through its own fast lane (thick: hand-off path and ingress check). Dotted items are still open; the bracket marks what a single-Vector PE omits. *Prototype status:* one 16-lane core; second core, multiply-add lanes, exp, top-k filter, sidecar write-back and reject/error reporting not yet in RTL.

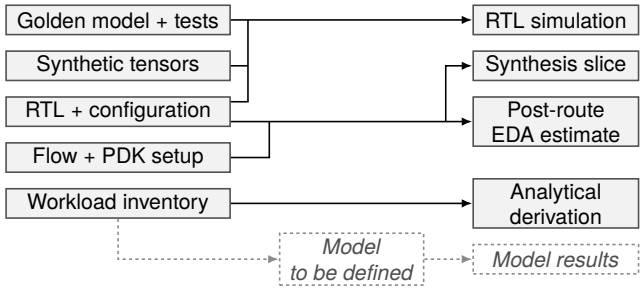


Fig. 8. Planned evaluation flow. Left: inputs. Right: the form of evidence that labels each number. Workload inventories are derived analytically and do not pass through a model. Dotted parts are open: the model has not been chosen, and only one of its inputs is drawn. The numerical front end that runs the synthetic tensors is not drawn.

B. Metrics, Operating Regimes and K

[sec5.2-p1] We report latency and throughput at three scopes: the end-to-end period of the complete chain, the steady-state throughput of useful results, and the local period of the hand-off and consumption stage. A local gain does not imply an end-to-end gain, and end-to-end parity does not show equal hand-off cost. The other metrics are area and timing, SRAM transactions, numerical error against the common error budget, and energy as action counts times sourced unit costs plus leakage, left open where a cost has no source. Two operating regimes are covered: L2-resident, with operands already in local storage, and supply-limited, where the PE port bounds their delivery [TBD-D4: definition of the supply-limited regime at the planned port width]. The total K of a GEMM is kept apart from the K_c of one tile, and placements are compared over the range of K that is legal for all of them [TBD-D7: legal range of K under the target operand loader].

TABLE V
SHARE OF MACS PER DECODED TOKEN BY OPERAND DATA TYPE OF THE REFERENCE IMPLEMENTATIONS (ANALYTICAL DERIVATION).

Operands; share at 8K / 1M (%)	V4-Pro	V4-Flash	V4.1-Flash
4 × 4 bit (index scoring)	0.9 / 42.5	2.4 / 64.3	1.2 / 38.3
4 × 8 bit (routed experts)[a]	44.2 / 15.9	44.0 / 9.3	47.3 / 29.5
8 × 8 bit (projections, shared expert)	34.0 / 12.3	29.9 / 6.3	30.6 / 19.1
BF16 × cache (attention core)	9.7 / 25.3	7.8 / 16.8	9.0 / 5.6
BF16 × BF16 (output projection)[b]	7.5 / 2.7	9.8 / 2.1	7.5 / 4.7
FP32 (remaining matrices)	3.6 / 1.3	6.2 / 1.3	4.4 / 2.8

[a] 4-bit weights. [b] Stored 8-bit. V4.1-Flash at 1M: its report's count.

C. Numerical Protocol

[sec5.3-p1] Numerical order is frozen before performance is compared (Section III-A), down to the final consumption point. A bit-accurate reference (golden) model defines the hardware semantics; all placements get the same tensors and seeds. The inputs are synthetic: we cannot run the models of Section V-D and do not use their weights. Shapes come from the workload's matrix products; activations and cache entries follow the quantization rules of the reference implementation, which gives none for weights [TBD-E3: value distributions and weight quantization rule of the synthetic tensors]. Quality means numerical error against a reference-precision result, not a task metric.

D. Workload: What Is Counted and How

[sec5.4-p1] The workload is the decode step of the three models of Section II-D [7], [8] [ref: DeepSeek-V4-Flash model repository]; V4-Flash is counted with the V4-Pro code under its own configuration. After prefill the reference implementation feeds one token per step, so counts are per sequence and per new token at context length L. MACs are counted only inside matrix products, other operations separately.

[sec5.4-p2] The counts are an analytical derivation; no model is run. Each matrix product contributes rows times inner-product length times layers, and other operations follow the loop bounds in the source. Operand types follow the reference code (Table V); index scoring is counted as 4-bit by 4-bit as its authors describe it, although the code computes it in BF16. For V4.1-Flash at 64K and above, re-scoring layers are counted over the candidate pool, as in its report [9]; the reference code scores every entry, which adds 58.8% to the MACs per token at 1M.

E. Model Fidelity and the Source of Performance Data

[sec5.5-p1] Single-PE latency comes from RTL simulation; the fused placement and everything beyond one PE need a model [TBD-P: source of performance data]. Whichever model is used, its results are labeled as such and the workload inventory is an input, not a result. A model calibrated against RTL is calibrated on PROTOTYPE, then on CORRECTED with new held-out points; reproducing a known fault of PROTOTYPE is agreement with the RTL, not a performance sample.

F. Physical Implementation

[sec5.6-p1] OpenROAD-flow-scripts, an open-source flow, implements the RTL on the ASAP7 predictive 7-nm PDK [ref: ASAP7 PDK; open-source implementation flow]. Area and timing come from synthesis and place-and-route runs under the stated library, constraint and SRAM-model assumptions; each is a post-route EDA estimate, not a silicon measurement. Runs use a single process corner, and the liberty views of the SRAM macros are generated: their zero leakage and zero minimum period are gaps in the model, not zero cost. Power is not reported until an activity-based method exists.

VI. RESULTS

[sec6-p0] Only Section VI-A and Section VI-F carry results today; the other subsections state the experiment, its figure (Fig. 9 for the main comparison) and how it will be read.

A. Workload Characterization

[sec6.1-p1] By analytical derivation, the operation mix of one decode step inverts as context grows (Fig. 2, Table I): at 8K feed-forward experts hold over half of the MACs and index scoring 0.9–2.4%; at 1M index scoring is the largest component. Only index scoring and, in V4-Pro and V4-Flash, the attention core of the layers that keep one entry per 128 tokens grow with L, so MACs per token rise 2.77×, 4.75× and 1.60× from 8K to 1M (V4-Pro, V4-Flash, V4.1-Flash).

[sec6.1-p2] At long context the most frequent non-GEMM operations follow index scoring and the attention core, one per 128 MACs of either in every model (Table I). Only index scoring multiplies two 4-bit operands (Table V). In the reference implementations, scales sit at matrix inputs and in caches while matrix outputs are scalar, so 1.39–3.31 million elements per token are dynamically quantized to 8 bits. How many of these a block-scaled hand-off removes, and which workload operations the consumer tiers of Section III-C stand for, is open [TBD-E4: quantization steps removed; mapping of workload operations to consumer tiers].

B. Correctness, Fail-Visibility and Numerics

[sec6.2-p1] Three checks precede any cost comparison (Fig. 10). (a) Whole-PE arithmetic closure: rows that a complete chain writes to L2 are read back and compared with the bit-accurate reference; we state whether output block exponents are covered [TBD-E1: whole-PE arithmetic closure, RTL simulation]. (b) Fail-visibility: every reachable class of mismatch, such as a tampered scale epoch or a result overwritten before consumption, must end in an error state with its credit settled exactly once [TBD-E2: fault-injection outcomes, RTL simulation]. (c) Numerics: we report the error of each placement against a reference-precision result on the synthetic tensors of Section V-C [TBD-E3: numerical quality on synthetic inputs; definition of the escape metric]. *Prototype status:* the GEMM-to-residual-add chain is verified in segments and, as a whole PE, only for completion.

C. When Does It Pay Off

[sec6.3-p1] The main experiment compares the five placements of Table II across the consumer tiers of Section III-C, both operating regimes and K (Fig. 9). S-E appears as an optimistic bound and a calibrated estimate [TBD-E4: result of the placement sweep, RTL simulation].

[sec6.3-p2] The sweep is read against hypothesis A (Section II-C), at each of the three scopes. A boundary, if one exists, is stated in three quantities: whether fusion is feasible, how long the result lives, and the granularity of metadata transactions (Section VI-D). Regimes in which the block form does not pay off are marked in Fig. 9, and the absence of a crossover is reported as such. *Prototype status:* no placement has been compared; until the 128 KiB L2 is in RTL, the primary comparison of Section III-A rests on a model.

D. Metadata Transactions and Storage Cost

[sec6.4-p1] Hypothesis B is tested with a transaction account computed analytically before any mechanism is built (Fig. 11). For each scale sidecar layout we count the six cost items of Section III-F for B-M and S-M. We report the storage that scale metadata occupies and whether a crossover exists. A bounded merge of the updates to one scale sidecar row is claimed as a mechanism only if it moves the crossover where the write-combining baseline cannot [TBD-E5: metadata transaction account].

E. The Cost of a Fail-Visible Hand-off

[sec6.5-p1] The cost of fail-visibility is reported as increments over the bare B-D hand-off (Fig. 12). We add, in order, the check at the consumer's ingress, aggregation of run-time errors at the command processor, commit of a payload with its scale metadata, and the second hop, where a stored result is read back and consumed. Each increment is charged its implementation cost, a post-route EDA estimate [TBD-E8: additive cost breakdown]. Checking at dispatch and checking

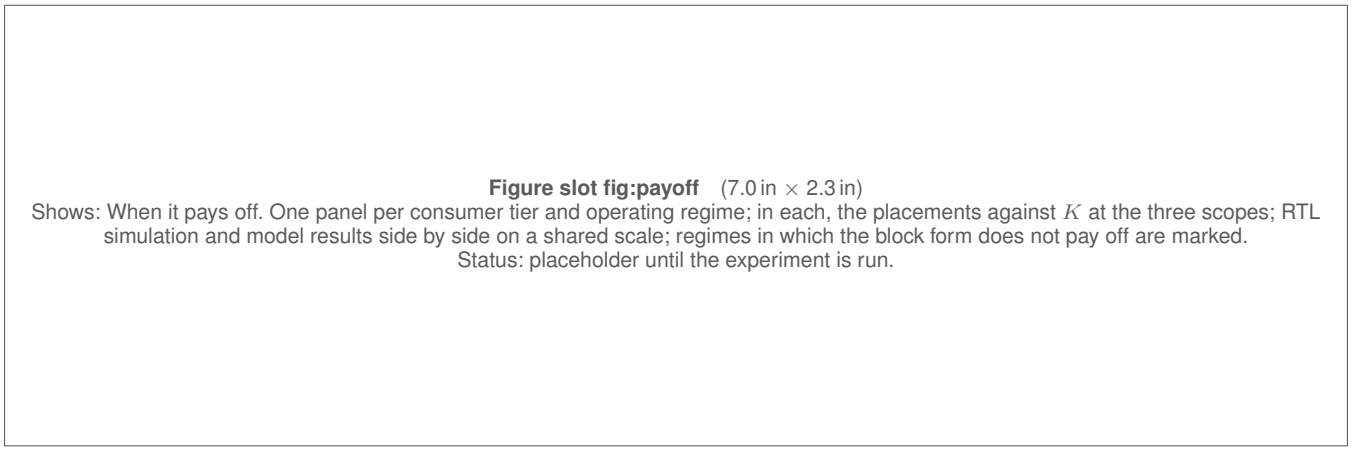


Fig. 9. Where keeping the block form pays off and where it does not: placements by consumer tier, operating regime and K at the three scopes of Section V-B; RTL simulation and model results in separate panels on one scale.

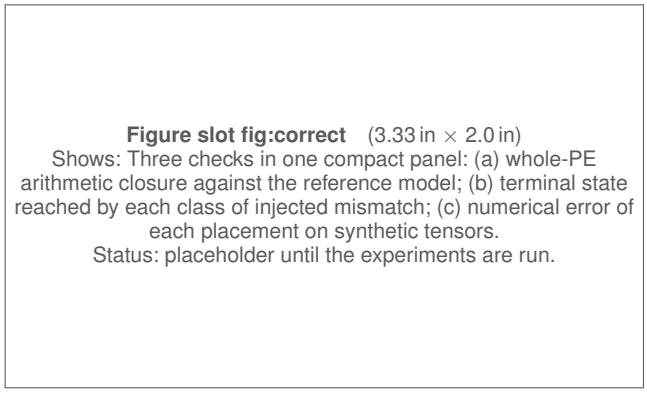


Fig. 10. Correctness (a), fail-visible behavior (b) and numerics (c): the three checks of Section VI-B.

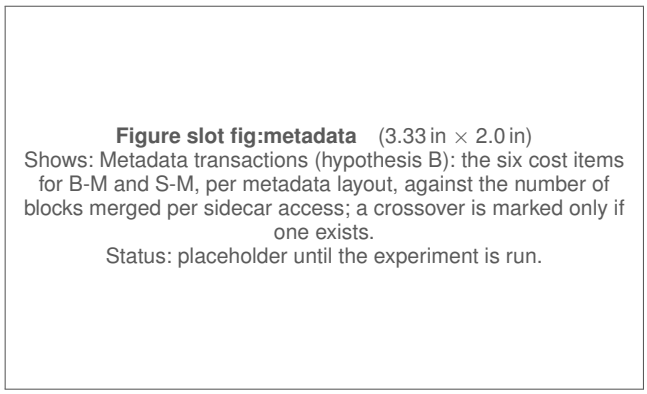


Fig. 11. Metadata transaction account for hypothesis B: cost per element of B-M and S-M against the number of blocks merged per scale sidecar access, per metadata layout. A crossover is marked only if one exists.

in the datapath are compared conditionally: both variants share the object table, leases and publication rule and differ only in where and how often the check runs.

F. Implementation

[sec6.6-p1] Fig. 13 shows the largest design that has a complete post-route layout today, the whole PE on ASAP7;

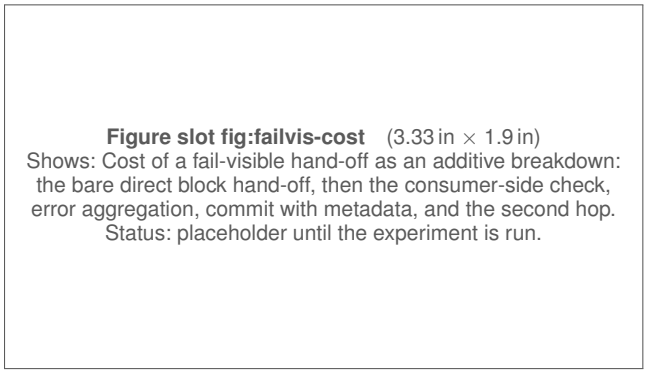


Fig. 12. Cost of a fail-visible hand-off, added step by step to the bare B-D path.

TABLE VI
PLACE-AND-ROUTE RUNS: THE PE OF FIG. 13, AND ITS TWO UNITS RUN ON THEIR OWN (ASAP7, POST-ROUTE; INTERMEDIATE VERSIONS).

Run	Die side (µm)	Standard cells[a]	Cell area[a] (µm^2)	SRAM macros	Clock constr.[b] (ps)
PE (Fig. 13)	1428.5	3,434,117	397,242	33	12,000
Tensor unit[c]	903.2	3,067,762	352,109	12	1,670
Vector unit[c]	330.5	265,898	28,560	0	1,670

Dies are square. [a] Logic cells only (no fill or tap cells); area is the sum of the cells' library footprints. [b] Clock period given to the tools as a constraint in the flow configuration; not an achieved period, and no timing result is reported here. [c] Block-level runs of the same RTL blocks; not crops of the figure. Figure and table are reissued after each full-flow run.

Table VI adds block-level runs of its two units. In the image, grey marks where standard cells sit, not how densely; the design is placed flat, so the frames are annotations, and the DMA engine and the fast-lane FIFO are too small to frame. The three runs use the same RTL, which precedes a numerical fix to the sum-of-squares unit, and differ in clock constraint and flow recipe. None of the three runs closes timing; the routing violations that remain in the whole-PE run are listed with the released sources. Per-component area and timing for

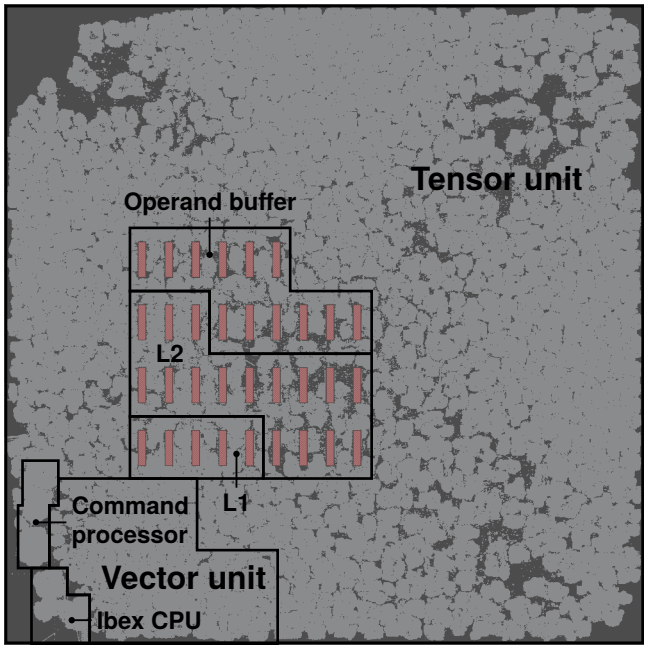


Fig. 13. Post-route layout of one PE (ASAP7 predictive PDK; an EDA result, not silicon; intermediate version), drawn from the routed design data. Frames mark SRAM macros (L1, L2, Operand buffer) or most of a block's cells; the rest is the Tensor unit. The RTL of this run predates the Vector sum-of-squares fix and has one Vector core and a 32 KiB L2; run data are in Table VI.

Figure slot fig:system-proj (3.33 in × 2.0 in)
Shows: System-level projection: whether a hand-off benefit found on one PE persists once transfer, publication to other PEs and workload mapping are added.
Status: placeholder; the source of performance data is not fixed.

Fig. 14. System-level projection. Every point is a model result, not RTL simulation; the model is not yet fixed.

CORRECTED and the target design are pending [TBD-E8: per-component breakdown].

G. System-Level Projection

[sec6.7-p1] The last question is whether a benefit found on one PE persists once transfer between PEs, publication to other PEs and workload mapping are added (Fig. 14). PROJECTED answers it with a model (Section IV-A); no throughput, latency or bottleneck is given [TBD-P: source of performance data] [TBD-E10: system-level projection]. If the model cannot close a transaction chain across PEs, the result becomes a bounded sensitivity analysis.

TABLE VII
CLOSEST PRIOR WORK, GROUPED BY MECHANISM: WHAT EACH LINE OF WORK ANSWERS AND WHAT IT LEAVES OPEN.

Answers	Leaves open
Block formats beyond GEMM: DB-Attn, MXInt ViT, MASE, BBAL block formats can cross the matrix/non-matrix boundary	when the block form pays off between separate, reused units
Production hand-offs: Blackwell with today's practice: S-E, S-D, and S-M plus packing	cuBLAS, MTIA 2i, Tenstorrent when the placement changes cost and numerics
Tags, write combining: CHERI, tagged tokens, Triggered Instructions atomic tag–data binding; tag check at the consumer; merged writes	multi-row numerical objects re-read under a lease; their cost

VII. RELATED WORK

[sec7-p1] **Block formats beyond the matrix engine.** Prior work has shown that block formats can cross the matrix/non-matrix operator boundary, in operator-specific engines. DB-Attn shares exponents between block floating-point matrix multiplication and softmax, with a softmax engine in RTL [17]. MXInt ViT maps every operator of a vision transformer into the MXInt domain [ref: MXInt ViT], and MASE generates dataflow accelerators whose inter-operator tensors stay in MXInt [ref: MASE]. BBAL adds a nonlinear unit in its BBFP format, but only after converting its inputs back from FP16 [12]. We ask instead when keeping the block representation pays off between separate, reused units, comparing placements as configurations of one RTL.

[sec7-p2] **Production hand-offs and the fused epilogue.** Three practices surveyed in Section II-B correspond to placements that we compare, in the order of Table VII: S-E, S-D, and S-M plus packing. Our comparisons stay between placements on one RTL and never cross process nodes.

[sec7-p3] **Data and metadata consistency, tags and write combining.** CHERI binds a tag atomically to its data and clears it on an ordinary write, per capability [ref: CHERI ISA specification]; it does not cover multi-row numerical objects read again under a lease after a statistic is computed. Tag comparison in tagged-token dataflow and Triggered Instructions is a precedent for checking at the consumer [ref: tagged-token dataflow; Triggered Instructions]. Write combining is common practice and our strong baseline for hypothesis B (Section III-F), not a contribution. The commit, lease and publish rules (Section III-D, Section IV-E) belong to our engineering mechanism: we report its cost and do not claim that correct metadata handling is itself new.

VIII. LIMITATIONS

[sec8-p1] Physical results are post-route EDA estimates on the ASAP7 predictive PDK at one process corner, with generated SRAM timing models; we claim no silicon, sign-off or frequency. Each result names its L2 storage contract; we never pair performance under the behavioral dual-port model with area and timing under single-port macros. The text describes the target design; Section IV states what the prototype lacks.

[sec8-p2] System-level results are model projections; calibration on one PE does not transfer to the network, shared storage or off-chip memory. Workload numbers are analytical counts of one decode step of one model family, not measurements. Numerical experiments use synthetic tensors: they report numerical error under stated distributions and say nothing about model quality. The same holds where the target design departs from the reference implementation: it quantizes the attention-core query to 8 bits (reference: BF16), and its online softmax and top-k selection may differ from the reference kernels. Fusion is evaluated only with a scalar epilogue, not in the block domain. The third consumer tier is a minimal instance, not a complete RMSNorm.

IX. Future Work

[sec9-p1] 3D stacking and a 3D mesh extend the 2D mesh assumed here. The SRAMs have no error detection or correction; adding it awaits a wider choice of memory macros or another process. The PE has no debug module, since simulation does not need one. The mesh-level Control Core, not evaluated here, could later be an out-of-order core, and larger meshes could use one core per control domain.

[sec9-p2] The mesh has one PE type and reserves capability bits in idle reports and task descriptors; a heterogeneous mesh of two PE types and its task mapping come later.

X. Conclusion

[sec10-p1] This paper asks when a GEMM result should stay block-scaled across the matrix-to-vector hand-off of a PE, rather than cross as a scalar or be consumed in a fused epilogue, and what a fail-visible hand-off costs. The placement sweep and the cost breakdown will give the answer in three quantities: whether fusion is feasible, how long the result must live, and the granularity of metadata transactions [TBD-E4: written after the results]. That answer is bounded by its evidence: numerical error on synthetic inputs, placements compared within one RTL, and physical costs as post-route EDA estimates, not silicon measurements.

Acknowledgment

[TBD-W: acknowledgments, to be written by the authors]

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Open placeholders in this draft

This list is generated from the `\TBD` marks in the text and is not part of the paper. Each line gives the placeholder id, the paragraph id and page where it occurs, and what it waits for. Marks of class W (text still to be written) are not listed.

TBD-D1 abs-p1, p. 1: form, width and division of labor of the two Vector cores
TBD-E4 abs-p1, p. 1: where the block form pays off and where it does not
TBD-E8 abs-p1, p. 1: implementation cost
TBD-P abs-p1, p. 1: system-level result; source of performance data
TBD-FIG fig:handoff, p. 1: concept figure, not drawn yet
TBD-REF sec1-p3, p. 1: MXInt ViT
TBD-REF sec1-p3, p. 1: MASE
TBD-REF sec1-p4, p. 1: DeepSeek-V4 paper
TBD-REF sec1-p4, p. 1: DeepSeek-V4-Flash model repository
TBD-N sec1-p5, p. 2: final wording; whether to name transaction granularity
TBD-P sec1-p6, p. 2: source of cycle-level and system-level performance data
TBD-E4 sec1-p7, p. 2: boundary of the pay-off region; no experimental result yet
TBD-D7 sec2.1-p2, p. 2: how the three axes read in the target design
TBD-REF sec2.2-p1, p. 2: Trainium3; Maia 200
TBD-REF sec2.2-p1, p. 2: Intel AMX
TBD-FIG fig:workload, p. 3: source plot exists; to be re-laid out for one column
TBD-REF sec2.4-p2, p. 3: DeepSeek-V4 paper
TBD-D1 sec3.1-p1, p. 3: whether the placements are compared on one Vector core or on both
TBD-N sec3.1-p2, p. 3: deciding metric and error budget, frozen before the sweep
TBD-D7 sec3.1-p2, p. 3: rounding at each narrowing point, to be re-read from the RTL once merged
TBD-E4 sec3.3-p1, p. 3: whether the third tier is evaluated, and at which sizes
TBD-D1 tab:placements, p. 4: one core or both
TBD-E4 tab:placements, p. 4: RTL or model
TBD-FIG fig:retain, p. 4: only a sketch exists
TBD-D7 sec3.4-p1, p. 4: layout of the written object, pending the write-back design
TBD-D7 sec3.5-p1, p. 4: where the encoding is stored, when a lease ends
TBD-D7 sec3.6-p1, p. 4: layout of the scale metadata, pending the write-back design
TBD-REF sec4.1-p1, p. 5: Ibex
TBD-D4 sec4.1-p1, p. 5: channels and handshake of the PE port, pending the network design
TBD-D3 sec4.1-p1, p. 5: bandwidth from shared storage into the mesh
TBD-D3 sec4.1-p2, p. 5: whether network endpoints and shared storage enter RTL
TBD-D5 sec4.2-p1, p. 5: which input modes map to each tier
TBD-D6 sec4.2-p2, p. 5: timing of this overhead, not yet evaluated
TBD-D1 sec4.3-p1, p. 5: buffering, credit return and check position of the two fast lanes
TBD-D1 sec4.4-p1, p. 5: form, width and division of labor of the two Vector cores
TBD-D7 sec4.5-p1, p. 6: L2 contract and write-back design open
TBD-E4 sec4.6-p1, p. 6: whether the scalar-through-storage placement is built in RTL or modeled
TBD-N sec5.1-p1, p. 6: neutral names for the three configurations
TBD-D7 sec5.1-p1, p. 6: final list of fixes, pending the merge plan
TBD-D4 sec5.2-p1, p. 7: definition of the supply-limited regime at the planned port width
TBD-D7 sec5.2-p1, p. 7: legal range of K under the target operand loader
TBD-E3 sec5.3-p1, p. 7: value distributions and weight quantization rule of the synthetic tensors
TBD-REF sec5.4-p1, p. 7: DeepSeek-V4-Flash model repository
TBD-P sec5.5-p1, p. 8: source of performance data
TBD-REF sec5.6-p1, p. 8: ASAP7 PDK; open-source implementation flow
TBD-E4 sec6.1-p2, p. 8: quantization steps removed; mapping of workload operations to consumer tiers
TBD-E1 sec6.2-p1, p. 8: whole-PE arithmetic closure, RTL simulation
TBD-E2 sec6.2-p1, p. 8: fault-injection outcomes, RTL simulation
TBD-E3 sec6.2-p1, p. 8: numerical quality on synthetic inputs; definition of the escape metric
TBD-E4 sec6.3-p1, p. 8: result of the placement sweep, RTL simulation
TBD-E5 sec6.4-p1, p. 8: metadata transaction account
TBD-E8 sec6.5-p1, p. 8: additive cost breakdown
TBD-FIG fig:payoff, p. 9: placeholder until the experiment is run
TBD-FIG fig:correct, p. 9: placeholder until the experiments are run
TBD-FIG fig:metadata, p. 9: placeholder until the experiment is run
TBD-FIG fig:failvis-cost, p. 9: placeholder until the experiment is run
TBD-FIG fig:system-proj, p. 10: placeholder; the source of performance data is not fixed
TBD-E8 sec6.6-p1, p. 10: per-component breakdown
TBD-P sec6.7-p1, p. 10: source of performance data
TBD-E10 sec6.7-p1, p. 10: system-level projection
TBD-REF sec7-p1, p. 10: MXInt ViT
TBD-REF sec7-p1, p. 10: MASE
TBD-REF sec7-p3, p. 10: CHERI ISA specification
TBD-REF sec7-p3, p. 10: tagged-token dataflow; Triggered Instructions
TBD-E4 sec10-p1, p. 11: written after the results